



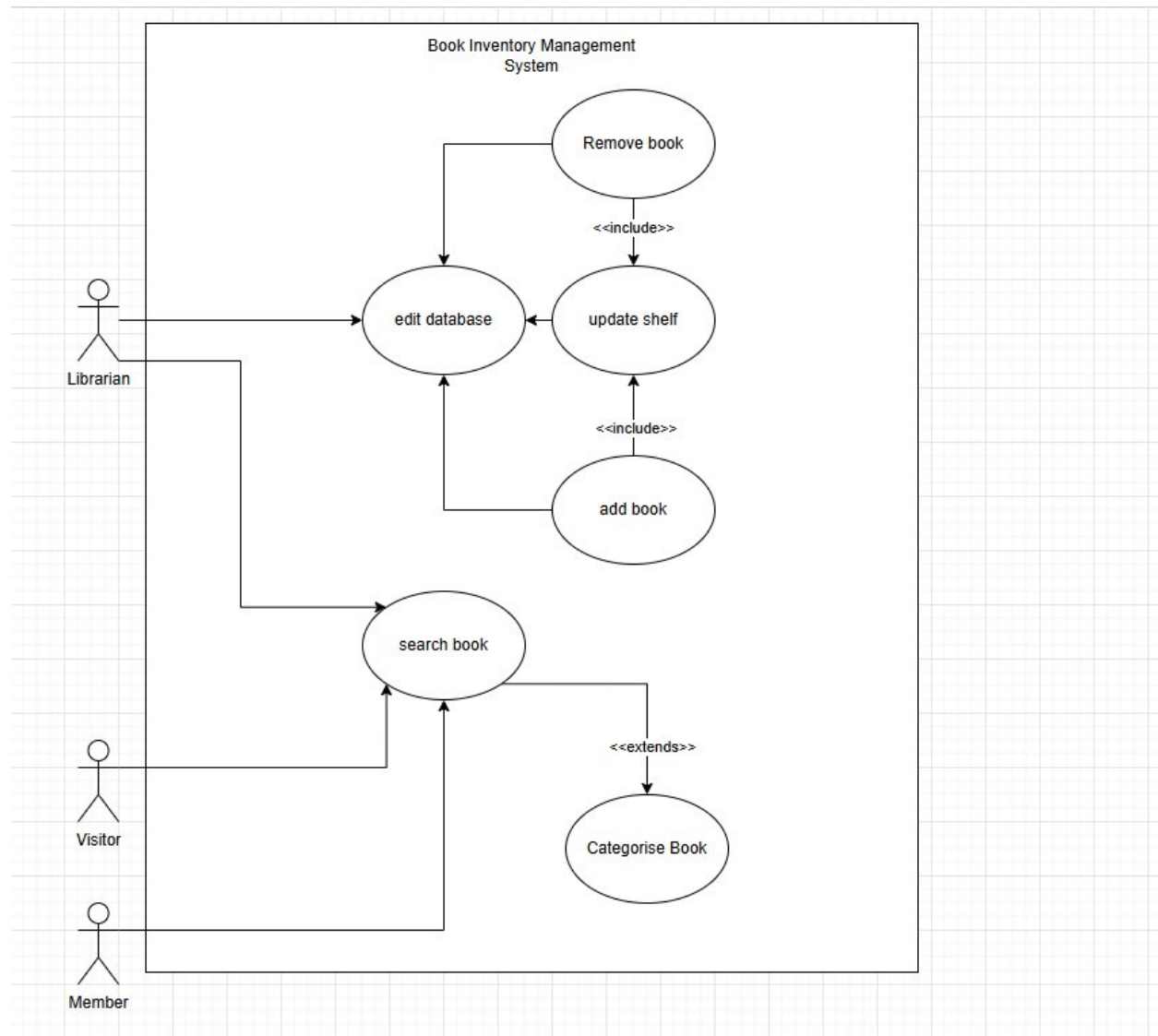
BCS 2163 : Object Oriented Design and Implementation

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ASSESSMENT : ASSIGNMENT (INDIVIDUAL)

Evo Smart Library		
GROUP : 1		
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USE CASE



USE CASE DESCRIPTION

Use Case Name	Suggest Shelf
Brief Description	Used by admin to get a shelf ID to assign a book
Actor	Librarian
Pre- Condition	User is logged in, Information about book is typed in
Basic Flow	1. System searches database for a shelf that is not full 2. System will display “Suggest Shelf” 3. User selects the suggestion 4. The suggested Shelf will be assigned to the book.
Alternative Flow	1. User uses a filter to pick which part of the building they want 2. System displays a list 3. User selects one selection from the list 4. The selected Shelf will be assigned to the book.
Exception Flow	1. User enters a specified shelf 2. The shelf will be assigned to the book.
Post Condition	• Book will be assigned a shelf.by the system or the user
Constraints	The system must only suggest shelves that currently have available space.

Use Case Name	Categorize Book
Brief Description	Helps users to categorize when searching for a book
Actor	Librarian, visitor, Member
Pre- Condition	Book has to exist in the book database
Basic Flow	1. User applies filter to the search bar 2. System will list every book that corresponds to the filtered information 3. User selects the from the list
Alternative Flow	1. User applies more than one filter to the search bar 5. System displays a list 6. User selects one selection from the list
Exception Flow	1. User selects a category. 2. System processes the request. 3. An unexpected system error occurs. 4. System displays an error message and terminates the search process.
Post Condition	• Search List will be categorized by user
Constraints	Users can only select categories that exist in the system.

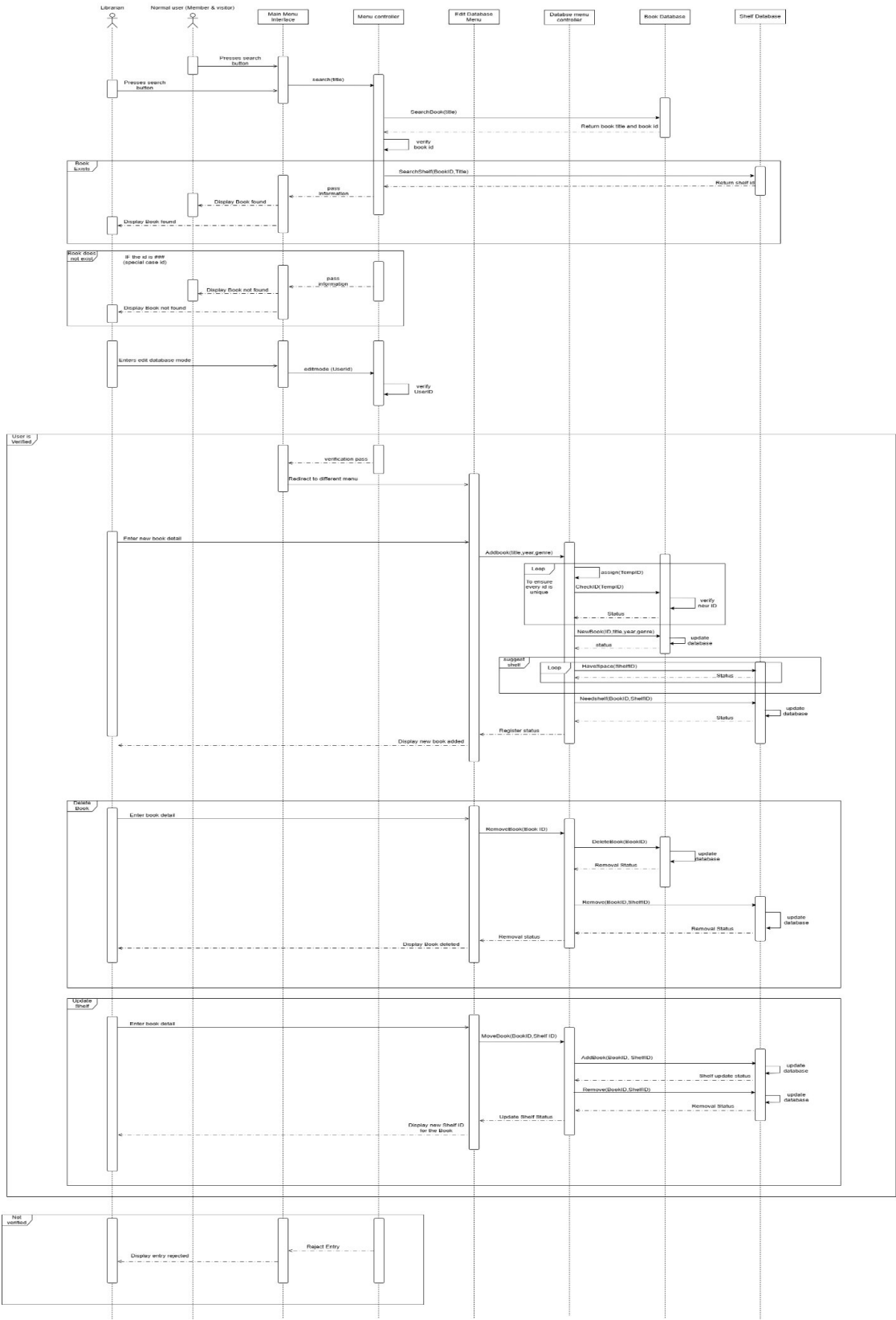
Use Case Name	Search Book
Brief Description	Used by users to search information about a book
Actor	Librarian, Member, Visitor
Pre- Condition	Book has to be entered in the database
Basic Flow	5. User enter the title or keyword of a book to find 6. System searches database matching the description 7. System will display “Book Found” 8. User selects view detail
Alternative Flow	5. User enters a keyword that matches several books 6. System displays a list of matching books. 7. User selects one book from the list 8. System displays the selected book details.
Exception Flow	1. User enter the title of a book to find 2. System searches database matching the description 3. System will display “Book not found”
Post Condition	• Book information is retrieved from the book database.
Constraints	The system can only show books that exist in the database

Use Case Name	Add Book
Brief Description	Allows the user to add a new book to the database
Actor	Librarian
Pre- Condition	User has to be logged in as admin
Basic Flow	1. User will enter book details 2. System will add information of the new book to the database 3. System will assign a shelf to the new book
Alternative Flow	1. System detects an existing copy of the book. 2. System increases the number of available copies instead of creating a completely new record. 3. System assigns a shelf
Exception Flow	1. User will enter book details 2. System will display an error message 3. System will cancel the operation
Post Condition	• New record of a book is made in the database
Constraints	Every Book has to have a unique ID

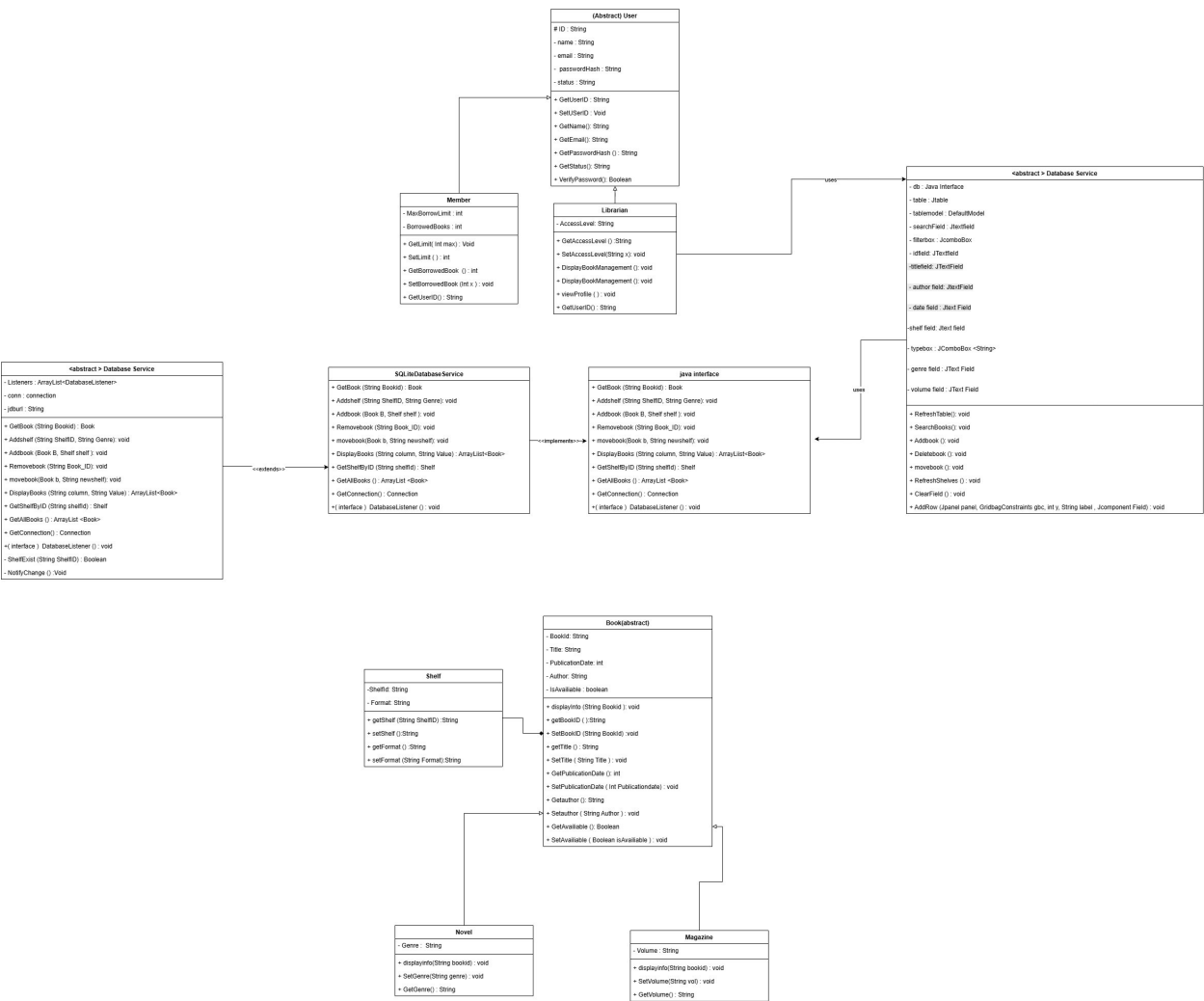
Use Case Name	Remove Book
Brief Description	Allows the user to remove a new book from the database
Actor	Librarian
Pre- Condition	User has to be logged in
Basic Flow	1. User will enter book title 2. System will verify information the information given 3. System will remove the record with the information from the database 4. System will update the shelf database
Alternative Flow	1. System detects an existing copy of the book. 2. User will enter number of books to remove 3. System decreases the number of available copies 4. System will update the shelf database
Exception Flow	1. User will enter book details 2. System will display an error message 3. System will cancel the operation
Post Condition	An old record is removed in the database
Constraints	Book has to exist in the database

Use Case Name	Update Shelf
Brief Description	Allows the user to move rearrange books in the database
Actor	Librarian
Pre- Condition	User has to be logged in
Basic Flow	1. User will enter book title 2. System will search the database for the book 3. User will enter the new shelf to assign to the book 4. System will update the shelf database
Alternative Flow	1. User clicks bulk select 2. User enters number of books to move 3. User will enter number of books to remove 4. System will update the shelf database
Exception Flow	1. User will enter book details 2. System will display an error message 3. System will cancel the operation
Post Condition	A book will be assigned a new shelf
Constraints	Book has to exist in the database

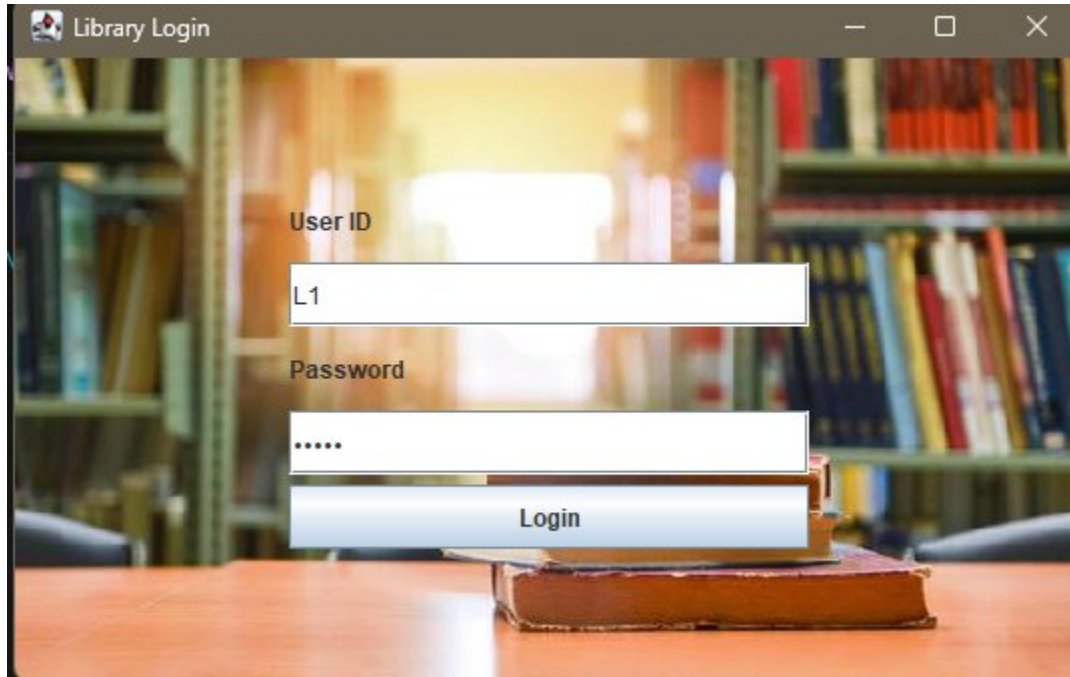
SEQUENCE DIAGRAM



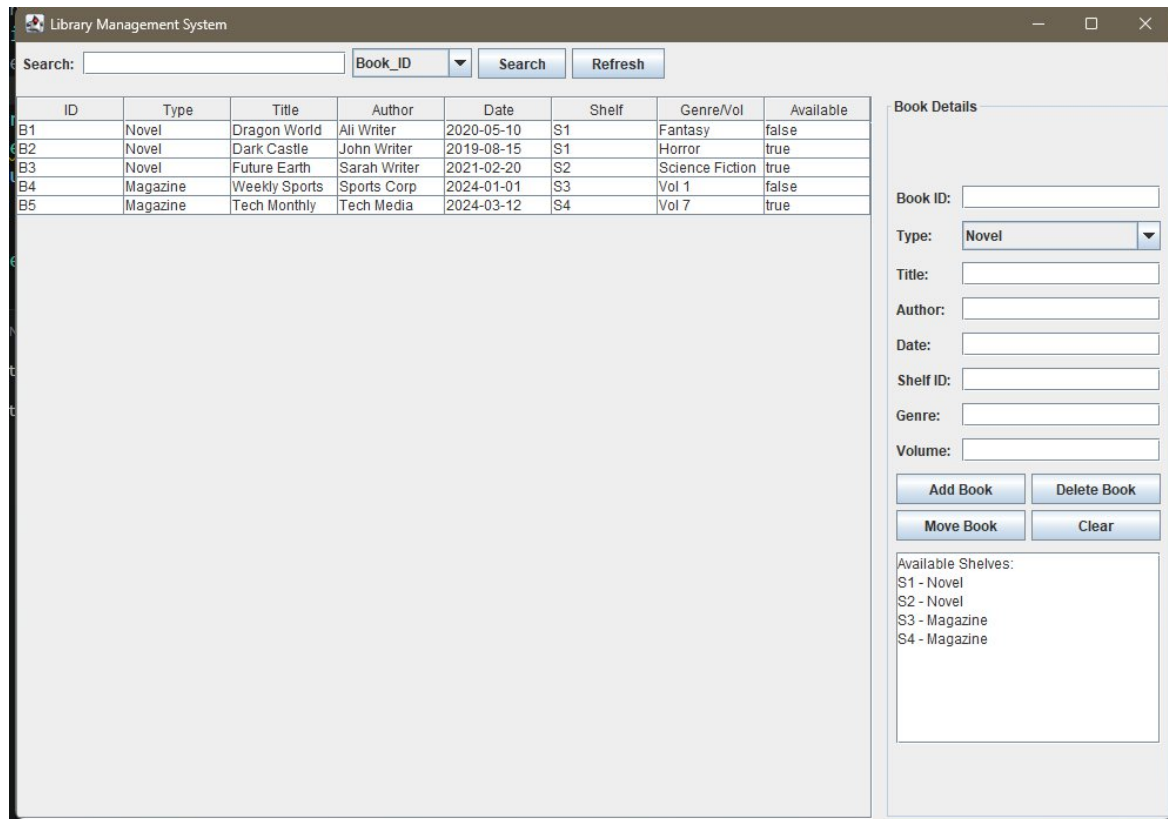
CLASS DIAGRAM



USER INTERFACE DESIGN



A screenshot of a 'Library Login' window. The window has a title bar with a small icon and the text 'Library Login'. The background is a blurred image of a library with bookshelves. The login form consists of two text input fields: the first is labeled 'User ID' and contains the text 'L1'; the second is labeled 'Password' and contains five dots. Below these fields is a blue 'Login' button.



A screenshot of a 'Library Management System' window. The window has a title bar with a small icon and the text 'Library Management System'. The interface includes a search bar with a dropdown menu set to 'Book_ID', and 'Search' and 'Refresh' buttons. Below the search bar is a table with 8 columns: ID, Type, Title, Author, Date, Shelf, Genre/Vol, and Available. The table contains 5 rows of data. To the right of the table is a 'Book Details' section with input fields for Book ID, Type (a dropdown menu set to 'Novel'), Title, Author, Date, Shelf ID, Genre, and Volume. Below these fields are four buttons: 'Add Book', 'Delete Book', 'Move Book', and 'Clear'. At the bottom of the 'Book Details' section is a text area labeled 'Available Shelves:' containing a list of shelves: S1 - Novel, S2 - Novel, S3 - Magazine, and S4 - Magazine.

ID	Type	Title	Author	Date	Shelf	Genre/Vol	Available
B1	Novel	Dragon World	Ali Writer	2020-05-10	S1	Fantasy	false
B2	Novel	Dark Castle	John Writer	2019-08-15	S1	Horror	true
B3	Novel	Future Earth	Sarah Writer	2021-02-20	S2	Science Fiction	true
B4	Magazine	Weekly Sports	Sports Corp	2024-01-01	S3	Vol 1	false
B5	Magazine	Tech Monthly	Tech Media	2024-03-12	S4	Vol 7	true

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